ALEKSANDAR RADULOVIĆ

Robot Learning · Reinforcement Learning · Embodied AI · Control Theory

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EDUCATION

2023 – PRESENT

Faculty of Technical Sciences
University of Novi Sad

- BSc, Computer and Control Engineering
- GPA: 9.85 / 10 — top 5 of class
- 4th year

TECHNICAL SKILLS

Programming: Python, C, C++, Assembly (basic)

ML & Robotics: PyTorch, NumPy, Pandas; VLA fine-tuning (SmolVLA); reinforcement & model-based RL; NVIDIA Isaac Sim

Tools: Linux, Git, 3D modeling & printing

LANGUAGES

Serbian: Native

English: C1 in progress

German: B1

PROFILE

I am a fourth-year Computer and Control Engineering student at the Faculty of Technical Sciences in Novi Sad, working at the intersection of machine learning and control theory to build intelligent, autonomous robotic systems. My core focus is Reinforcement Learning for continuous control — an interest I first pursued through research in model-based RL, and have since extended to autonomous driving challenges. In parallel, I have become increasingly drawn to generative models — diffusion and flow matching — as a principled foundation for learning robot policies, the same class of methods that underpins modern Vision-Language-Action systems. To build this foundation rigorously, I completed MIT's flow matching course (6.S184), while the EEML summer school broadened my research experience more generally. Being shortlisted for the highly competitive EPFL Research Internship further validated my potential to contribute to cutting-edge research. My long-term ambition is to become a top-tier researcher in Embodied AI and Robot Learning, and I am eager to join environments with strong mentorship where I can fully dedicate myself to data-driven continuous control and expanding the impact of ML in the physical world.

RESEARCH EXPERIENCE

Research Intern → Continuing Researcher — 2026 – Present

The Institute for Artificial Intelligence Research and Development of Serbia (IVI), Novi Sad

- Building an end-to-end Vision-Language-Action (VLA) pipeline on a FANUC CRX collaborative arm: designed and 3D-printed a custom end-effector, collected teleoperated demonstrations, and synchronized camera and joint-state streams into training data.
- Recreated the full workcell in NVIDIA Isaac Sim to automate large-scale demonstration generation, then fine-tuned and evaluated SmolVLA policies on the physical robot, investigating the sim-to-real gap.
- Invited to continue after the two-month internship; now improving the VLA model and extending it to new manipulation tasks alongside my studies.

Research Poster — TD-MPC Reproduction — 2026

EEML Summer School 2026

- Reproduced TD-MPC (model-based RL) from scratch in a CleanRL-style codebase and ran an ablation on the latent-consistency loss, trading stability for a faster, higher-peaking policy.
- Awarded Best Poster among ~250 participants, the majority of them MSc and PhD students.

SELECTED PROJECTS

Hardware-in-the-Loop Simulation of a Sorting Plant — 2026

- Built a real-time HIL simulation in LabVIEW on NI CompactRIO using a producer–consumer architecture with state machines and SCADA integration.

Pedestrian Trajectory Prediction — 2025

- Predicted pedestrians' near-future motion on the ETH BIWI dataset using engineered relative-motion and social (nearest-neighbour) features; compared linear models, Random Forest and XGBoost (best: 0.19 m ADE) and served the model via a FastAPI app.

Road-User Detection with YOLOv8 2025

- Trained and compared YOLOv8 Nano/Small and a focal-loss class-balanced variant for detecting cars, pedestrians and cyclists on a custom ~400-image dataset, reaching 80% mAP@50 on the test split.

PUBLICATIONS & POSTERS

- *Temporal Difference – Model Predictive Control: A Reproduction Study* — research poster, EEML Summer School 2026 (Best Poster Award).

HONORS & AWARDS

- **Best Poster Award** — EEML Summer School 2026 (~250 participants).
- **1st place, Elektrijada** (national student competition) — Mathematical Analysis 1.
- **1st place, Elektrijada** (national student competition) — Mathematical Analysis 2.
- **1st place, EEStech Challenge — Autonomous Vehicles** (Novi Sad); advancing to the international round.
- **Shortlisted, EPFL Excellence Research Internship** — 2026.

CERTIFICATES & COURSES

- **Generative AI with Flow Matching (MIT 6.S184)**
  Completed — ODE/SDE solvers, conditional flow matching, classifier-free guidance.
- **Structured & Advanced Structured Programming (C)**
  Foundation „Centar za mlade talente".
- **Object-Oriented Programming (C++)**
  Foundation „Centar za mlade talente".

INDEPENDENT STUDY

- Modern Robotics: Mechanics, Planning, and Control (Lynch & Park).
- Reinforcement Learning (D. Silver, UCL / DeepMind lecture series).
- Diffusion and flow-matching methods for robot policy learning.